

Ancestral skies staffing

By Harry Hawkins 48019675

Sets

S Skills
 J Tasks
 I Technician
 D Days

Data

e_{is} score of technician $i \in I$ for skill $s \in S$
 k_j skill required for job $j \in J$
 t_j time required to complete task $j \in J$
 p_j number of technicians required to complete task $j \in J$
 a_j title description of task $j \in J$
 M maximum time available for each technician: 36 hours a week
 Md maximum time available for each technician on day $d \in D$: 10 hours a day
 b bonus score after training: 7

Variables

X_{ij} Binary variable of technician $i \in I$ allocated to task $j \in J$
 W_{id} Binary variable of technician $i \in I$ working on day $d \in D$
 Y_{is} Binary variable representing if technician $i \in I$ receives training for skill $s \in S$
 T_i Time allocated to technician $i \in I$ on day $d \in D$
 L_j Total skill allocated to task $j \in J$
 H_{id} Hours worked by a technician $i \in I$ on day $d \in D$

Objective

Optimise the skill scores to complete all tasks

$$\max \sum_{j \in J} L_j$$

Constraints

$$X_{ij} \in 0, 1 \quad \forall i \in I, j \in J \quad (1)$$

$$Y_{is} \in 0, 1 \quad \forall i \in I, s \in S \quad (2)$$

$$T_{id} = \sum_{j \in J} t_j \cdot x_{ij} \quad \forall i \in I \quad (3)$$

$$T_{id} \leq M \quad \forall i \in I \quad (4)$$

$$L_j = \sum_{i \in I} (e_{ik} \cdot x_{ij} + Y_{ik} \cdot b) \quad \forall j \in J, k \in k_j \quad (5)$$

$$\sum_{i \in I} x_{ij} = p_j \quad \forall j \in J \quad (6)$$

$$X_{3j} \neq X_{8j} \quad \forall j \in J \quad (7)$$

$$X_{6j} \neq X_{7j} \quad \forall j \in J \quad (8)$$

$$X_{3j} \neq X_{12j} \quad \forall j \in J \quad (9)$$

$$X_{10j} \neq X_{8j} \quad \forall j \in J \quad (10)$$

$$H_{id} \leq Md \cdot W_{id} \quad \forall i \in I, d \in D \quad (11)$$

$$\sum_{d \in D} W_{id} \leq 5 \quad \forall i \in I \quad (12)$$

$$X_{ij} + X_{in} \leq 1 \quad \forall i \in I, j \in J, n \in \{j \mid a_j = \text{Nightly telescope operation}\} \quad (13)$$

$$\sum_{i \in I, s \in S} Y_{is} = 5 \quad (14)$$

$$\sum_{s \in S} Y_{is} \leq 1 \quad \forall i \in I \quad (15)$$

The first two constraints (1, 2) specify the binary variables used in the model where (2) is used for applying the additional training scores. Constraint (3) refers to the total time worked for worker i . Constraint (4) applies the limit of work hours for each worker. The constraint applied to L_j specifies how the total skill for each job. As seen in (5) it is the efficiency of worker i in the skill required for job j with the addition of a skill bonus if the worker i had training. The next constraint (6) ensures there is enough technicians to do job j . From (7) to (10) ensure that certain colleagues do not work together. Constraint (11) connects the binary variable W and hours worked on day H such that if $W_{id} = 0$ then $H_{id} = 0$ and if $W_{id} = 1$ then H_{id} can be up to the maximum hours a day Md . The next constraint (12) ensures workers only work for a maximum of 5 days a week. Constraint (13) causes any worker who is rostered on the night shift to have that as their only job for that day. Finally (14) and (15) refer to the bonus efficiencies given to those who have training, the first constraint limits the number of workers who have training to 5, and the second states that each of the 5 workers chosen must be unique.

Client Report

Ancestral skies staffing

In an attempt to keep this report concise, weekly scores are used to indicate the rostering. In the appendix tables showing the most effective rostering for each day produced by the final model are shown.

Communication 6

Given the set of tasks needing to be completed for the week and the time it takes to complete them a total skill score achieved by the end of the week was found. With the application of the constraint that technicians could only work up to 36 hours a week and each task specifying the required amount of technicians to complete the job, the score was 2676.9.

Communication 7

To follow the kinship avoidance practices, additional constraints were put in such that the technician IDs given in the table below did not work together on a job. This reduced the total skill score to 2615.4, a decrease of 61.5.

Technician	3 & 8	6 & 7	3 & 12	8 & 10
------------	-------	-------	--------	--------

Communication 8

Shifting into the framework from weekly tasks to daily tasks with the addition of a maximum work load of 10 hours a day for each technician, the new score for the week was 2530.4. There was once again a decrease as previously some technicians with higher skills worked more than 10 hours a day, but this constraint needs to redistribute the time to each technician such that some jobs may no longer get the optimal technician for the job. To further examine the effect of adding time restrictions to daily work, I observed the total skill score that could be achieved if you extend the hours up to 11 or down to 9 hours. The scores can be seen in the table below.

Maximum daily work hours	8	9	10	11	12
Weekly skill score	infeasable	2522.3	2530.4	2535.4	2539.4

As you can see in the table above, increasing the maximum hours a technician can work a day does increase the total skill score for that week but only by at most 5. This may not be worth the increase if it increases costs of the project. On the other hand if we decrease the daily hours we find that the score decreased by 7.1 for 9 hours. Also we can see that 9 hours a day is the minimum amount of time required to complete all tasks described by the data as the 8 hour limit returned a infeasable model.

Communication 9

With the additional constraint of technicians rostered to do "Nightly telescope operation" required not to do any tasks before hand on the same day, the score was once again reduced to 2489.3. Similarly to the previous communication, this decrease is due to losing some efficiency from other jobs due to those who work the night shift not being chosen for any day shifts.

Communication 10 - Final Model

An assumption was made when incorporating the additional weekly training which was each technician can only receive one training session each. This means, of the five technicians receiving training they will all be unique and only increase in one skill only. By giving additional training to five technicians and increasing their score in one skill by seven, this drastically improves the weekly score to 2979.3. The technicians you should train and the skills they should improve are indicated in the table below.

Technician	Skill
5	Environment monitoring
6	Mechanical maintenance
7	Environment monitoring
10	Environment monitoring
12	Software operation

However to give you an indication on whether a technician should be allowed to have multiple training sessions, the total weekly score would remain 2979.3 with technician 9 receiving 3 sessions. As there is no change in the result either model could be appropriate, but I will submit the model with the assumption stated at the beginning of the section as it aligns more with the spirit of Ancestral Skies supplying training and education to the community as indicated in your briefing statement.

Summary

It is worth noting that the model given only keeps track of the first week of applying this technicianing. Although the roster does not need to change to maintain the weekly score of 2979.3, it could be improved if training at the beginning of each week continues and technicians scores keep increasing. The final rostering can be seen in the appendix below.

Appendix

Mon	Technicians
Electronics diagnostics and repair has technicians	[0, 9]
Real-time fault troubleshooting has technicians	[0, 9]
Assembly and testing of mechanical fixtures has technicians	[2, 12]
Monitoring weather and system status has technicians	[6]
Mentoring junior technician has technicians	[12]
IT and network maintenance has technicians	[1, 3]
Updating and reviewing system logs has technicians	[0, 10]
Nightly telescope operation has technicians	[7, 11]
Instrument calibration has technicians	[4, 6]
Motorized component adjustments has technicians	[12]
Preparing dome and telescope for night operations has technicians	[9]
Optical alignment checks has technicians	[6, 8]
Safety tests (interlocks and motion limits) has technicians	[4]
Instrument data-acquisition monitoring has technicians	[1]
Adaptive optics system validation has technicians	[2]
Documenting issues and proposed improvements has technicians	[10]
Assisting astronomers with instrument setups has technicians	[2]
Hardware installation and integration has technicians	[3, 5]
Testing remote-observing systems has technicians	[1, 6]
Preventive maintenance on control systems has technicians	[4]
Cryogenic system checks has technicians	[0, 8]
Sensor and actuator testing has technicians	[0]
Cleaning optical components has technicians	[5, 8]
Facility maintenance (HVAC, water, power) has technicians	[2, 12]
Tues	Technicians
Preventive maintenance on control systems has technicians	[4, 5]
Instrument calibration has technicians	[4]
Nightly telescope operation has technicians	[3, 11]
Monitoring weather and system status has technicians	[10]
Sensor and actuator testing has technicians	[0]
Electronics diagnostics and repair has technicians	[0, 9]
Drives and dome troubleshooting has technicians	[2, 12]
Motorized component adjustments has technicians	[12]
Hardware installation and integration has technicians	[5, 7, 8]
Real-time fault troubleshooting has technicians	[0]
Instrument data-acquisition monitoring has technicians	[9]
Preparing dome and telescope for night operations has technicians	[9]
Adaptive optics system validation has technicians	[2]
Cryogenic system checks has technicians	[0, 7]
Facility maintenance (HVAC, water, power) has technicians	[12]
Documenting issues and proposed improvements has technicians	[10]
Testing weather-monitoring systems has technicians	[7, 10]
Optical alignment checks has technicians	[8]
IT and network maintenance has technicians	[8, 9]

Wed	Technicians
Monitoring weather and system status has technicians	[6]
Electronics diagnostics and repair has technicians	[0, 1]
Software updates and testing has technicians	[7]
Structural and mechanical repairs has technicians	[2, 3]
Instrument data-acquisition monitoring has technicians	[1]
Cryogenic system checks has technicians	[0, 7]
Mentoring junior technician has technicians	[4]
Assembly and testing of mechanical fixtures has technicians	[2, 3]
Testing remote-observing systems has technicians	[1, 7]
Instrument calibration has technicians	[4, 6]
Testing weather-monitoring systems has technicians	[6]
Thermal management of dome and telescope has technicians	[6]
Nightly telescope operation has technicians	[9]
Preparing dome and telescope for night operations has technicians	[7]
Coordinating instrument upgrades has technicians	[4]

Thur	Technicians
Drives and dome troubleshooting has technicians	[0, 12]
Adaptive optics system validation has technicians	[2]
Assisting astronomers with instrument setups has technicians	[4]
Instrument data-acquisition monitoring has technicians	[9]
IT and network maintenance has technicians	[8]
Mentoring junior technician has technicians	[4]
Cryogenic system checks has technicians	[0, 8]
Monitoring weather and system status has technicians	[6]
Electronics diagnostics and repair has technicians	[9]
Real-time fault troubleshooting has technicians	[0, 9]
Instrument calibration has technicians	[2, 4]
Safety tests (interlocks and motion limits) has technicians	[4]
Testing weather-monitoring systems has technicians	[6, 10]
Nightly telescope operation has technicians	[3, 11]
Optical alignment checks has technicians	[8, 12]
Preparing dome and telescope for night operations has technicians	[9]
Documenting issues and proposed improvements has technicians	[10]
Assembly and testing of mechanical fixtures has technicians	[12]

Fri	Technicians
Updating and reviewing system logs has technicians	[10]
Preparing dome and telescope for night operations has technicians	[9]
Motorized component adjustments has technicians	[3]
Documenting issues and proposed improvements has technicians	[10]
Electronics diagnostics and repair has technicians	[9]
Assembly and testing of mechanical fixtures has technicians	[2, 3]
Instrument calibration has technicians	[2, 4]
Monitoring weather and system status has technicians	[6]
Instrument data-acquisition monitoring has technicians	[1]
IT and network maintenance has technicians	[8, 9]
Safety tests (interlocks and motion limits) has technicians	[4]
Real-time fault troubleshooting has technicians	[9]
Nightly telescope operation has technicians	[5, 11]
Testing weather-monitoring systems has technicians	[6, 10]
Structural and mechanical repairs has technicians	[1, 2, 3]
Thermal management of dome and telescope has technicians	[6]

Sat	Technicians
Drives and dome troubleshooting has technicians	[0, 12]
Thermal management of dome and telescope has technicians	[6]
Cryogenic system checks has technicians	[0]
Instrument data-acquisition monitoring has technicians	[1]
Documenting issues and proposed improvements has technicians	[0]
Cleaning optical components has technicians	[8]
Mentoring junior technician has technicians	[12]
Assisting astronomers with instrument setups has technicians	[12]
Testing remote-observing systems has technicians	[1, 6]
Nightly telescope operation has technicians	[7]
Optical alignment checks has technicians	[5, 8]
Monitoring weather and system status has technicians	[6]
Instrument calibration has technicians	[5, 6]
Preventive maintenance on control systems has technicians	[5]
Preparing dome and telescope for night operations has technicians	[5]

Sun	Technicians
Assisting astronomers with instrument setups has technicians	[12]
Monitoring weather and system status has technicians	[10]
Preparing dome and telescope for night operations has technicians	[5]
Nightly telescope operation has technicians	[11]
Documenting issues and proposed improvements has technicians	[10]
Real-time fault troubleshooting has technicians	[1]
Electronics diagnostics and repair has technicians	[1]
Mentoring junior technician has technicians	[12]
Instrument data-acquisition monitoring has technicians	[1]
Safety tests (interlocks and motion limits) has technicians	[5]
Software updates and testing has technicians	[7]